Course work II Data Structures and Algorithms.

Attempt the work in groups of 5 according to the class groups and submit this work on the platform individually

Problem 1: The Scenario: You are a software developer at a local logistics company that uses a large fleet of Boda-Bodas (motorcycle taxis) for urgent city deliveries in Kampala. Your goal is to develop a system to find the most efficient route for a driver who has multiple drop-off points, minimizing travel time (which is heavily affected by unpredictable traffic and road conditions).

Part A: Algorithm Design

a) Design a pseudocode algorithm for finding the *optimal* route that a Boda-Boda driver can take to visit all specified delivery locations in the shortest time. Explain the core graph algorithm (e.g., Dijkstra's, A*, or an approach to the Traveling Salesperson Problem/TSP variant) and the key data structures used to manage the route search space.

b) Describe how your algorithm would handle common Kampala traffic constraints, such as:

i. Peak-hour gridlock: Roads where travel time significantly increases at certain times.
ii. Road closures/Flooding: Temporarily inaccessible routes due to construction or weather.
iii. Toll/Fee points: Points that add a monetary cost (which might be an alternative metric to minimize).

Part B: Data Structures

a) Explain which specific data structures would be most efficient to store and manipulate information about the road network of a city like Kampala (e.g., intersections as nodes, roads as edges) and the delivery locations. Justify your choices based on typical graph operation efficiency (adding, deleting, and querying edges/nodes).

b) Discuss the space-time trade-off between using an Adjacency Matrix versus an Adjacency List to represent the road network. Which one would you recommend for a city like Kampala,

and how does that choice impact the memory usage and the execution speed of your route-finding algorithm?

Problem 2: E-Commerce and Artisan Recommendation System for Local Products: The Scenario: You are working on a Ugandan e-commerce platform, "MusawoMarket," focused on connecting local artisans (selling crafts, fabrics, and farm produce) directly with buyers. You need to implement a personalized recommendation system to help users discover new products and support local businesses.

Part C: Recommendation Algorithm

a) Design a pseudocode algorithm for generating personalized product recommendations for a registered user. Explain the logic of your chosen recommendation approach (e.g., Collaborative Filtering, Content-Based, or Hybrid) and the data structures necessary to track and process user interactions and product features.

b) Describe how your algorithm incorporates the following uniquely local factors into the recommendation process:

i. Region/Tribe Preference: Users from a specific region might prefer crafts/produce from that area.
ii. Seasonality: Recommendations for fresh produce should prioritize items currently in harvest.
iii. New/Emerging Artisans: How do you ensure new, unrated local artisans get visibility without being drowned out by popular, established ones (the "cold start" problem)?

Part D: Data Structures

a) Discuss which data structures could be used to efficiently store and retrieve:

i. User-Item Ratings/Interactions (the history of purchases and clicks).
ii. Product Features (e.g., category, material, region of origin).

Provide reasons for your data structure choices (e.g., efficiency in lookups, insertions).